Joyce Xu | Great Neck South High School, NY | joycexu346@gmail.com | https://jjoycex.github.io| 631-308-7221

Research Projects and Publication

Gridshell Modeling | Hofstra University Summer Research *2025-2026*
Co-developed/authored 2 novel low-energy low-waste methods for fabricating small-scale gridshell model. Published in IASS-IWSS'26
Animatronic Hand| Science Research *2025-2026*
Developed a 3D-printed animatric hand system for real-time sign language integrated with on-device speech recognition model.
Smart Walking Stick | Science Research *2024-2025*
Developed a smart walking stick to detect obstacles with vocal alerts to aid visually impaired people.
Microgravity Neurophysiology in Drosophila | Science Research *2024-2025*
Investigated the effects of NAC on metabolic and neurophysiological stress in D. melanogaster with simulated microgravity.

Leadership

Science Olympiad | Co-President *Jun 2026 – Present*
Leading 45 member Science Olympiad team with duties in recruiting and training new members, proctor and grade tryout tests, evaluating team structure and strategy, working with team faculty on purchases and invitational competition selections.
Science Olympiad | Board Member and Treasurer *Sep 2024 – Jun 2026*
Handled club deposits and sales transactions in school records, held bake sales partnering with local organizations, proctored and graded tryout tests, helped team members on their events, discussed team logistics in board meetings.
Exit 33 (High school art and literary magazine) | Copy Editor *Sep 2023 – Present*
Editing short stories and poems being published in the Exit 33 magazine, contributing short stories to the magazine.

Awards

Research Publication
- Lee, M., Xu, J., Yeh, T., & Segal, E. M. (alphabetical order). "Methods for Creating Small-Scale Polylactic Acid (PLA) and Wood Gridshell Models." Peer-reviewed paper published in the *Proceedings of the IASS-IWSS 2026 Annual Symposium*, International Association for Shell and Spatial Structures, Turin, Italy, September 14–18, 2026.

Science Olympiad
- 2026 National Competition: 9th (Experimental Design), 15th (Robot Tour), 16th (Electric Vehicle), 29th (Helicopter)
- 2026 New York State Competition: 1st (Experimental Design), 2nd (Helicopter), 5th (Electric Vehicle), 8th (Robot Tour)
- 2023-2026 Invitational Competition awards from 1st to 10th place. Regional Competition awards from 1st to 9th place.

MIT Beaver Summer Program Awards
- 2026 BWSI Basic of ASICs Best Capstone Award recipient
- 2026 BWSI Basic of ASICs Assignment Tutorial Competition Award recipient
- 2026 BWSI Basic of ASICs Hype Video Competition Award recipient

Science Research
- 2026 Long Island Science and Research Fair (LISEF) Third Place
- 2026 WAC Lighting Invitational Science Fair Second Place
- 2025 WAC Lighting Invitational Science Fair Third Place
- 2025 SAAWA Science Research Mini Grant Awardee

Music
- 2025 Live performance at Kammermusiksaal of the Philharmonie in Berlin, Germany (piano)
- 2023 Live performance at Bosendorfer-Saal at Mozarthaus in Vienna, Austria (piano)
- 2025-2021 Grand Prize Virtuoso International Music Competition First Prize (piano)
- 2025-2023 AADGT International Young Musician Competition Gold Medal (piano)

Other Activities

- [Creative Writing] Iowa Young Writers' Studio summer program *2024*
- [Music] NYSSMA All-County Music Festival (first violinist) *2024 - 2026*
- [Music] Metropolitan Youth Symphony Orchestra (first violinist) *2023 - 2024*

Volunteers and Jobs

- Weekly English tutoring for fourth grade student *Sep 2025 - Apr 2026*
- Music Buddy violin mentor help younger students achieve 100 on NYSSMA Lvl 6 and MYO admission *2024 - Present*
- Play for autistic children at *United in Love* concert and S*pecial Love for Special You* concerts *2024 and 2019*